

Timing and speed are going to become vital for survival and success of the future organisations. No organisation in today's age can survive without agility and responsiveness to changing environments. Systemic efficiencies can only be brought in and improved through control and instrumentation. With companies becoming more and more complex and dispersed, there is need for efficient manpower.”

—Sanjay Mittal, managing director, Yogasa Systems

of the small-scale industry in India requires well-trained instrumentation engineers having knowledge of computers and instrumentation,” opines Prof. Rekha Agarwal, head of Department of Instrumentation and Control Engineering, Amity School of Engineering and Technology.

Instrumentation engineering is one of the complicated but sophisticated branches of engineering discipline that may be studied as a separate branch or along with electronics engineering. The study mainly focuses on the design, configuration and automated systems.

“It deals with measurement of various physical quantities like temperature, pressure, level, flow, speed, sound, light intensity and control of the same in various industries. Instrumentation system is widely used in industries, viz, automotive, pharmaceuticals, chemical, fertilizers, power plants, pollution control, biomedical, food processing, electronic product manufacturing and textile. With the advancement and widespread applications of electronics and computers in instrumentation and control, the syllabus is framed to include core courses of electronics as well as computer engineering,” explains Prof. R.D. Kokate, head of Department of Instrumentation Engineering,

MGM's Jawaharlal Nehru Engineering College.

According to Rohit Sinha, head-human resources, L&T Engineering, E&C Division, “The Indian automation market has acquired the critical momentum to propel the instrumentation and control industry to a higher growth trajectory. Instrumentation is a well-established technology, both in the manufacturing sector and infrastructure. He feels that India's hope of emerging as an economic superpower depends a lot on how we groom our engineers to leverage this technology. By transferring global-quality learning processes, we can convert a much larger percentage of the emerging manpower to more enriching careers.”

An instrumentation engineer can find a wide range of career opportunities in all sectors of industries

ranging from automotive to health-care. The passouts are absorbed in power plants, fertilisers and chemicals industry, petrochemicals industry, pharmaceutical industry, cement factories, healthcare services, consulting services, navigational and aerospace organisations, food processing industry and weather stations to name a few. Among the hundreds of corporate employers, some are Texas Instruments, HCL, TCS, ABB, Larsen & Toubro (L&T), National Instruments, Pepsi, Nagarjuna Fertilizers, Xilinx, Honeywell, Wockhardt, Bechtel and Saint-Gobin. In the public sector, there are SAIL, BHEL, NTPC, ONGC, Indian Oil, etc.

Know your role

Instrumentation and control engineers design, build and manage systems that are used in a range of industrial settings such as manufacturing, health-care, food processing, mining and energy production. “An instrumentation engineer is invariably required where there is an engineering activity,” says Kokate.

Keep in mind, as an instrumentation engineer, you have to monitor, measure, regulate and control physical quantities like pressure and temperature. Further, you may need to control product movement, actuators and positioning devices. Your main objective will be to ensure that the systems and processes operate effectively, efficiently and safely.

Most of the jobs available in this field can be bucketed under two broad categories. “An instrumentation en-

The shortage of instrumentation engineers becomes more due to less number of colleges offering B.Tech degree in instrumentation and control in India.”

—Prof. Rekha Agarwal, head of Department of Instrumentation and Control Engineering, Amity School of Engineering and Technology